

Delete from Vs Truncate

Delete from:

The DELETE command is used to delete specific rows from a table based on a condition.

advantages:

Deletes specific data (using WHERE) Can be undone within a transaction (ROLLBACK) Does not delete the table structure.

disadvantages:

Slower with large datasets
Logging records every deletion

TRUNCATE:

The TRUNCATE command is used to delete all data within a table at once without conditions.

Advantages:

Faster than DELETE
Deletes all data instantly
Does not use WHERE
Returns only the empty table

disadvantages:

Cannot delete specific rows
WHERE does not work
Cannot be easily undone in some cases
Resets identity (in SQL Server)

MERGE Statement in SQL Server

Introduction

The **MERGE** statement in SQL Server is used to synchronize data between two tables. It combines the functionality of **INSERT**, **UPDATE**, and **DELETE** into a single statement. **MERGE** is especially useful when data from a source table needs to be compared with a target table, and actions should be taken based on whether matching records exist.

Instead of writing separate **INSERT** and **UPDATE** statements, **MERGE** allows developers to perform these operations efficiently in one command, improving readability and maintainability.

```
MERGE TargetTable AS T
USING SourceTable AS S
ON T.ID = S.ID
```

```
WHEN MATCHED THEN
    UPDATE SET T.Name = S.Name
```

```
WHEN NOT MATCHED THEN
    INSERT (ID, Name)
    VALUES (S.ID, S.Name);
```

Explanation:

- **Target Table:** The table that will be updated.
- **Source Table:** The table containing new or updated data.
- **ON:** Defines the matching condition between the two tables.
- **WHEN MATCHED:** Executes when matching records are found.
- **WHEN NOT MATCHED:** Executes when no matching record exists in the target table.

Advantages of MERGE

1. Combines multiple operations into a single statement.
2. Reduces code complexity.
3. Improves performance when synchronizing large datasets.
4. Makes data integration and ETL processes easier.
5. Ensures consistency between source and target tables.

Conclusion

The **MERGE** statement is a powerful SQL feature that simplifies data synchronization by combining INSERT, UPDATE, and optionally DELETE operations into one command. It is widely used in database maintenance, reporting systems, and data warehousing environments where data must be kept consistent between multiple tables.